

Assignment 2 – Case Study Lab

Stage 2 Lab Activities

SmartCare Requirements Engineering

AI OFF -> AI ON -> VERIFY | 1 hour

Learning objectives

- Analyse the SmartCare client brief.
- Identify stakeholders and scope.
- Write functional and non-functional requirements.
- Develop user stories and Given-When-Then acceptance criteria.
- Use AI to critique requirements without allowing it to invent stakeholder needs.
- Produce SmartCare Requirements Specification v1.0.

Part A - Client Brief: AI OFF

SmartCare uses spreadsheets and paper records. Staff report duplicate bookings, difficulty finding patient information, inconsistent appointment status and limited appointment history. Management wants a small, maintainable patient, practitioner and appointment system.

Part B - Stakeholders and Scope: AI OFF

Identify at least four stakeholders. Create In Scope and Out of Scope lists. Label uncertain features as provisional rather than confirmed.

Part C - Functional Requirements: AI OFF

Write 8-12 numbered functional requirements using FR-01, FR-02 and so on. Each should describe one observable capability.

Part D - Non-Functional Requirements: AI OFF

Write 4-6 numbered non-functional requirements covering appropriate qualities such as reliability, maintainability, usability, data integrity or testability.

Part E - User Stories and Acceptance Criteria: AI OFF

Write 4-6 user stories. For at least three, create Given-When-Then acceptance criteria including one negative or failure scenario.

Part F - AI Requirements Review: AI ON

Prompt: Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Part G - VERIFY the AI Review

Classify each significant AI suggestion as Accepted, Modified, Rejected, or Unverified. Explain the evidence used.

Part H - Finalise SmartCare v0.2

Submit stakeholder analysis, scope, 8-12 FRs, 4-6 NFRs, 4-6 user stories, acceptance criteria, assumptions/open questions and selected AI review evidence.

Reflection

In 150-250 words: What did AI notice that you missed? What did AI invent or overreach on? Which requirement changed after review? Why must requirements have evidence?